

List of question of chapter 10

1. Define plaintext and cipher text with math execution (P-4)

◆ Plaintext:



Plaintext is the original, readable message or data that you want to protect. It's the input to the encryption process.

✎ Example:

HELLO

◆ Ciphertext:



Ciphertext is the unreadable, scrambled(ঠেলাঠেলি করিয়া একত্র করান) output of the encryption process. It looks like gibberish(অর্থশূন্য) and is meant to hide the real content from unauthorized users.

🛡 Example:

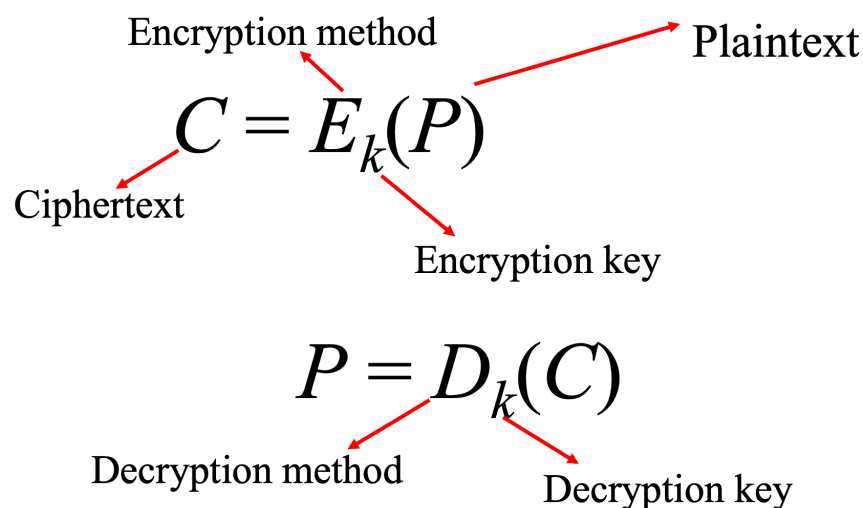
If we encrypt HELLO, it might become:

KHOOR

1 2 3 4 Mathematical Execution (Using Caesar Cipher as an Example)



Caesar Cipher is a basic encryption method where each letter is shifted by a fixed number of positions in the alphabet.



The encryption model (for a symmetric-key cipher)

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2. Definition of cryptography, cryptanalysis (codebreaking), cryptology

Cryptography

 Cryptography is the science of **securing communication** by converting readable data (**plaintext**) into an unreadable format (**ciphertext**) to prevent unauthorized access.

-  **Goal:** Ensure **confidentiality, integrity, authentication, and non-repudiation**.
-  **Example:** Encrypting a message like "HELLO" into "KHOOR" using Caesar Cipher.

Cryptanalysis (Codebreaking)

 Cryptanalysis is the art and science of **breaking cryptographic systems**, i.e., trying to decrypt data **without knowing the secret key**.

-  **Goal:** Discover the **plaintext** or the **key**.
-  **Example:** Figuring out that "KHOOR" was encrypted with Caesar Cipher and reversing it to get "HELLO".

Cryptology

 Cryptology is the **study of both cryptography and cryptanalysis**. It's the broad field that encompasses the **design of secure communication systems** and the **methods used to break them**.

-  Think of cryptology as the umbrella term:

Cryptology = Cryptography + Cryptanalysis

3. Compare active and passive intruder (p-6)

Comparison: Active vs Passive Intruder

Aspect	Active Intruder 	Passive Intruder 
Definition	Actively modifies, injects, or disrupts data	Only listens or monitors without altering
Goal	Cause harm, change messages, impersonate	Gather information secretly
Example Actions	Altering messages, replay attacks, DoS	Eavesdropping, traffic analysis
Impact	Direct and often visible	Indirect and stealthy
Detection	Easier to detect	Harder to detect
Defense Mechanisms	Authentication, firewalls, intrusion detection	Encryption, secure protocols

Examples

- **Active Intruder:** A hacker sends fake login pages to steal passwords.
- **Passive Intruder:** A hacker silently captures data packets over Wi-Fi.

Let me know if you'd like a visual diagram or real-world analogy too!

4. Give an example of substitution cipher (p -12)

Example of a Substitution Cipher: Caesar Cipher



A **substitution cipher** replaces each letter in the plaintext with another letter according to a specific rule.

Example Using Caesar Cipher (Shift = +3)

◆ Plaintext:

HELLO

◆ Encryption Rule:

Each letter is shifted **3 positions forward** in the alphabet.

Letter	H	E	L	L	O
Shift	K	H	O	O	R

◆ Ciphertext:

KHOOR

Decryption:

To decrypt, shift **3 positions backward**:

KHOOR → HELLO

This is a classic example of a **monoalphabetic substitution cipher**. Want to see an example with symbols or a polyalphabetic cipher like Vigenère next?

5. Example of Transposition cipher (p-15)

Example of Transposition Cipher



A **transposition cipher** does not change the characters themselves but **rearranges their positions** according to a specific pattern or key.

Key: MEGABUCK

(Each letter in the key corresponds to a column)

Plaintext (to be encrypted):

milliondollar

Step-by-step (Columnar Transposition)

1. **Assign numerical order to the key** (alphabetical order of letters):

M E G A B U C K
7 4 5 1 2 8 3 6

2. **Write the plaintext row by row under the columns:**

```
MEGABUCK
74512836
-----
milliond
ollarxxx
```

(We add x's to fill the empty cells)

3. **Read the columns in the order of the key (1 to 8):**

- 1 → A → l a
- 2 → B → i r
- 3 → C → n x
- 4 → E → i l
- 5 → G → l l
- 6 → K → d x
- 7 → M → m o
- 8 → U → o x

4. **Final Ciphertext (column-wise read):**

```
la ir nx il ll dx mo ox
```

 **Ciphertext:** lairnxillldxmoox

 **Note:** Transposition ciphers preserve the frequency of letters but scramble their positions.

 Decryption of the Transposition Cipher

 Recap of Encryption

- **Key:** MEGABUCK
- **Key Order (Alphabetical Index):**

```
MEGABUCK
74512836
```

Sorted key: A B C E G K M U

Numerical order for rearrangement:

```
A(1), B(2), C(3), E(4), G(5), K(6), M(7), U(8)
```

- **Ciphertext:** lairnxillldxmoox
- Split into chunks by column length: Each column has **2 letters**, so we divide into 8 parts of 2 letters each.

1. A → l a
2. B → i r
3. C → n x
4. E → i l
5. G → l l
6. K → d x
7. M → m o
8. U → o x

 Step-by-Step Decryption

1. **Arrange columns by original key order:**

MEGABUCK
74512836

Now place the values from sorted order (A to U) into their correct positions based on this map:

M(7)	E(4)	G(5)	A(1)	B(2)	U(8)	C(3)	K(6)
mo	il	ll	la	ir	ox	nx	dx

2. **Rebuild the rows** row-wise from the arranged columns:

- Row 1: m i l l i o n d
- Row 2: o l l a r x x x

3. **Rejoin rows to recover the plaintext:**

milliondollarxxx

Remove padding (x) if added just to fill the grid.

✅ **Final Decrypted Plaintext:**

milliondollar

Let me know if you'd like a **Python script** or **printable version** of this explanation!

6. Compare substitution cipher and transposition cipher

🔄 Comparison Table

Feature	Substitution Cipher	Transposition Cipher
Definition	Replaces each character or group with another	Rearranges positions of characters without changing them
Example	Caesar Cipher, Monoalphabetic Cipher	Columnar Transposition, Rail Fence
Effect on Letters	Letters are changed	Letters remain the same, only shuffled
Statistical Distribution	Changed (frequency of letters is altered)	Unchanged (frequency remains same)
Security Weakness	Can be broken using frequency analysis	Can be broken using pattern/position analysis
Key Type	Substitution table or shift value	Key usually defines column order or pattern
Encryption Nature	Symbol-for-symbol replacement	Permutation of characters
Usage	Often used for obscuring identity of symbols	Used to obscure order without changing identity

✅ **Conclusion:**

- **Substitution** = disguise characters
- **Transposition** = shuffle characters
- Both are used in **symmetric encryption** and can be **combined** for stronger security.

Let me know if you want a side-by-side visual too!

7. principle likhte hobe (1st diff: definition 2nd: statistical distribution)

7. Example of RSA algorithm (p:22 - p:24)

Example of RSA Algorithm

 Based on Slides P:22–P:24 of your document

✓ RSA Step-by-Step Example

Step 1: Choose two prime numbers

From the slide:

$$\begin{aligned}p &= 17 \\q &= 11\end{aligned}$$

Step 2: Compute modulus (n) and Euler's totient (z)

$$\begin{aligned}n &= p \times q = 17 \times 11 = 187 \\z &= (p - 1)(q - 1) = 16 \times 10 = 160\end{aligned}$$

Step 3: Choose decryption key (d)

Choose d such that $\text{gcd}(d, z) = 1$

Let $d = 7$ (since $\text{gcd}(7, 160) = 1$)

Step 4: Find encryption key (e)

Find e such that:

$$de = 1 \pmod{z}$$

(We will divide $de = 7e$ by $z = 160$

and the remainder will be 1 for example you can get $e = 23$)

Try $e = 23$:

$$7 \times 23 = 161 \equiv 1 \pmod{160}$$

✓ So, $e = 23$

 **Public Key = (e = 23, n = 187)**

 **Private Key = (d = 7, n = 187)**

✉ Encryption Formula

Encoded message, $C = P^e \pmod{n}$

Where:

- P is the plaintext number
- C is the ciphertext

🔓 Decryption Formula

Decoded Message $P = C^d \pmod{n}$

Where:

- C is the ciphertext
- P is the recovered plaintext

Mini Example (Slide 24)

To encrypt a message, P , compute $C = P^e \pmod{n}$. To decrypt C , compute $P = C^d \pmod{n}$.

Let us consider another example:

$$p = 3, q = 11$$

$$n = pq = 33, z = (p-1)(q-1) = 20$$

$d = 7$, since 20 and 7 does not have common factor

Now $7e = 1 \pmod{20}$ which provides $e = 3$

$C = P^3 \pmod{33}$ which provides encoded value C

Plaintext (P)		Ciphertext (C)		After decryption		
Symbolic	Numeric	P^3	$P^3 \pmod{33}$	C^7	$C^7 \pmod{33}$	Symbolic
S	19	6859	28	13492928512	19	S
U	21	9261	21	1801088541	21	U
Z	26	17576	20	1280000000	26	Z
A	01	1	1	1	01	A
N	14	2744	5	78125	14	N
N	14	2744	5	78125	14	N
E	05	125	26	8031810176	05	E

Sender's computation

Receiver's computation

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Fig. An example of the RSA algorithm.

✅ Message successfully encrypted and decrypted!

Let me know if you'd like help solving an RSA problem with your own values or converting this to Python code!

8. Compare conventional and digital sign (p:27-28)

🔑 Comparison: Conventional Signature vs Digital Signature

Aspect	Conventional Signature	Digital Signature
Inclusion(অন্তর্ভুক্তি)	Signature is part of the physical document	Sent as a separate file along with the message
Verification	Verified manually by comparing with a sample	Verified using algorithms and public key cryptography
Duplicity	Original and copy can be distinguished	No difference between original and copy unless timestamped
Relationship	One signature can be reused for multiple documents	Each signature is unique to a specific message

✅ Additional Insights:

- **Conventional Signature:** e.g., handwritten signatures in banks or official papers
- **Digital Signature:** Ensures **authenticity**, **integrity**, and **non-repudiation** using **encryption and hashing**

Let me know if you want a visual summary or printable table version!

9. Give the basic condition concept of message digest line(p:38 er 4 ta point)

📌 Basic Conditions of Message Digest

📄 From Slide P:38



A **message digest** is a **fixed-length hash value** generated from a variable-length message using a **one-way hash function**. It ensures **data integrity and security**.

✅ 4 Key Properties (Slide 38):

1. **Given P, it is easy to compute MD(P)**
→ The hash function should be **efficient and fast**.
2. **Given MD(P), it is effectively impossible to find P**
→ The function must be **one-way**, preventing reverse computation.
3. **Given P, no one can find P' such that MD(P') = MD(P)**
→ Ensures **collision resistance**.
4. **A small change in input (even 1 bit) results in a very different in the output of MD(P)**
→ Known as the **avalanche effect**.

These properties make message digests crucial in **digital signatures, data integrity checks, and authentication systems**.

Let me know if you want an example using SHA-1 or SHA-256!

10. Explain operation secure digital sign, under SHA algorithm(p:40)



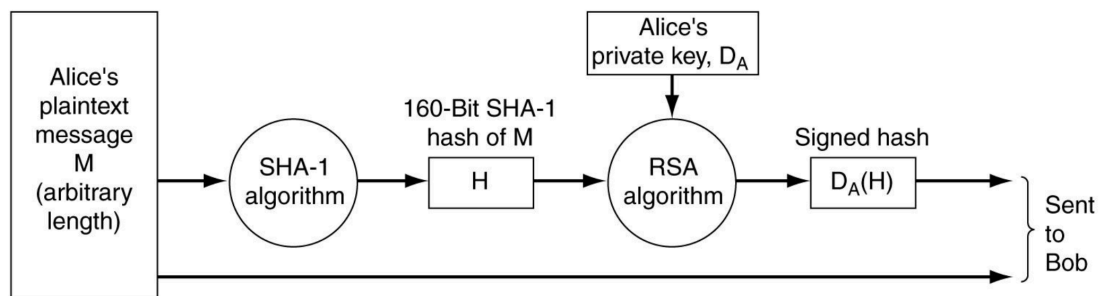
Secure Digital Signature Operation under SHA Algorithm



Based on Slide P:40

✅ Process Overview (Using SHA-1 and RSA)

Here Alice's plaintext message is fed into the SHA-1 algorithm to get a 160-bit SHA-1 hash. Alice then signs the hash with her RSA private key and sends both the plaintext message and the signed hash to Bob like fig. below.



Use of SHA-1 and RSA for signing nonsecret messages

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1. Alice's Message Preparation

- a. Alice writes a **plaintext message**.

2. Hashing with SHA-1

- a. The plaintext is passed through the **SHA-1** hash algorithm.
- b. This generates a **160-bit message digest (hash)**.

3. Signing the Hash with RSA Private Key

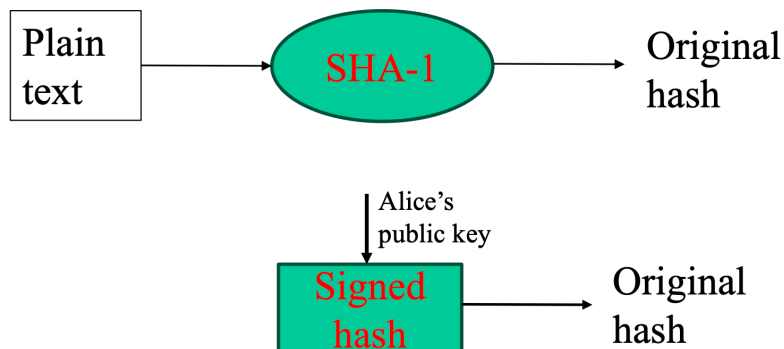
- a. Alice **encrypts the hash** using her **RSA private key**.
- b. This **signed hash = digital signature**.

4. Sending Data

- a. Alice sends **two things** to Bob:
 - i. The **original plaintext message**
 - ii. The **digital signature** (signed hash)

✓ Bob's Verification Process

After receiving the message, Bob computes the original SHA-1 hash from plain text and also applies, Alice's public key to the signed hash to get the original hash, H . If the two agree, the message is considered valid. Since there is no way for intruder to modify the message (plaintext) while it is in transit and produce a new one that hashes to H , Bob can easily detect any change.



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5. Recomputing the Hash

- a. Bob applies SHA-1 to the **received message** to get a **new hash**.

6. Decrypting the Signature

- a. Bob **decrypts the digital signature** using Alice's **public key**, recovering the **original hash**.

7. Validation

- a. Bob compares the **newly computed hash** with the **decrypted hash**.
- b. ☒ If they match → message is **authentic and unaltered**
- c. ☒ If they don't → message has been **tampered with**

Why It's Secure:

- Without Alice's **private key**, no one can forge the signature.
- The **SHA-1 hash ensures integrity**, and RSA ensures **authentication**.

Let me know if you'd like a diagram or code demonstration of this process!

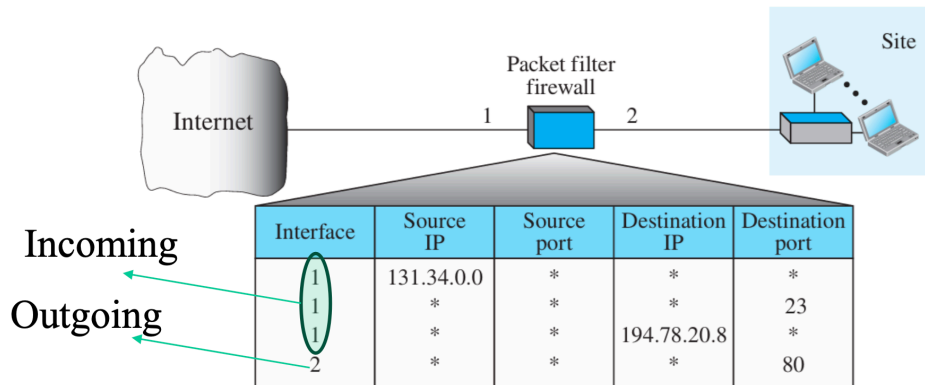
11. Mention 3 example of packet filtering on a firewall (P:55)

3 Examples of Packet Filtering on a Firewall

Based on Slide P:55

Packet-filter firewall

Figure below shows an example of a filtering table for packet filtering.



1. Incoming packets from network 131.34.0.0 are blocked (security precaution). Note that the * (asterisk) means “any.”
2. Incoming packets destined for any internal TELNET server (port 23) are blocked.
3. Incoming packets destined for internal host 194.78.20.8 are blocked. The organization wants this host for internal use only.
4. Outgoing packets destined for an HTTP server (port 80) are blocked. The organization does not want employees to browse the Internet.

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Here are three specific examples from the slide that illustrate **packet filtering rules** in a firewall:

✓ Example 1:

Block incoming packets from network 131.34.0.0

- Security precaution against a suspicious external network

✓ Example 2:

Block incoming packets destined for internal TELNET servers (port 23)

- Prevents external users from remotely logging into internal systems using TELNET

✓ Example 3:

Block incoming packets to internal host 194.78.20.8

- The organization reserves this host for internal use only

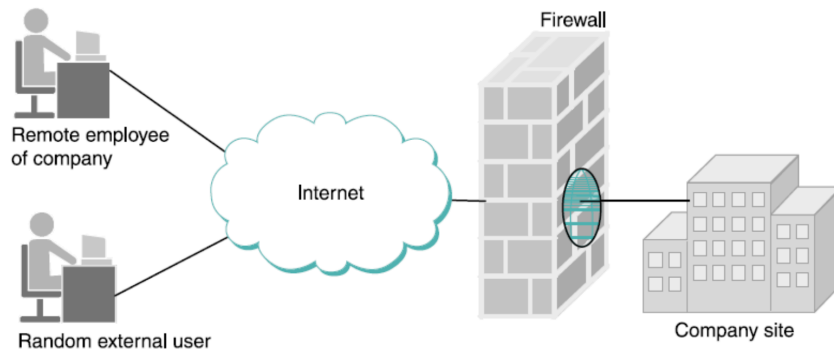
These packet filtering rules help **control traffic flow**, **enhance security**, and **prevent unauthorized access**.

Let me know if you want more firewall examples or a visual table!

12. what is firewall(p:52 first para)

🔥 What is a Firewall?

From Slide P:52 (First Paragraph)



A **firewall** is a combination of **hardware and software** that:

- **Isolates an organization's internal network** from the broader Internet
- **Controls the flow of network traffic**
- **Allows some packets to pass and blocks others** based on defined rules

✅ In short:

A firewall **filters packets** between a private network and the Internet to enhance **security** by **permitting or denying** data transmission.

Let me know if you'd like the types or examples of firewall next!

13. How app gateway firewall works(p:57)

An **Application Gateway Firewall** (also known as a **Proxy Firewall**) operates at the **application layer** of the network.

✅ **Working Mechanism:**

1. **Inspects the actual content** of the messages, not just headers or ports.
2. Makes filtering decisions based on:
 - a. **Header fields**
 - b. **Message size**
 - c. **Specific content/keywords**
3. Acts as an **intermediary** between the internal network and the outside world.
4. For example, a **mail gateway** can:
 - a. Examine every incoming or outgoing email
 - b. Allow or block based on specific words (e.g., "nuclear", "bomb")

🔑 **Key Point:**

It provides **deeper inspection** than packet-filter firewalls, improving control over **application-level threats**.

Let me know if you'd like a visual example or comparison with other firewall types!

14. Flow intruder can enter through app gateway firewall(P: 58)

👤 **How an Intruder Can Bypass an Application Gateway Firewall**

📄 *Based on Slide P:58*

Even with a well-configured **application gateway firewall**, **intruders can still find ways to get through**. Here's how:

🔒 **Common Intrusion Methods:**

1. **IP Spoofing (Fake Source Address)**

- a. An attacker can **forge a false source IP address** to trick the firewall into thinking the packet is from a trusted network.
 - b. This can **bypass filters** that allow only certain IPs.
-

2. Insider Threats

- a. A **trusted internal user** might intentionally leak sensitive data.
 - b. Example: Converting secret text into an image (e.g., .jpeg) and sending it out, **bypassing keyword filters**.
-

Key Point:

Even the most advanced firewalls **cannot prevent all threats**, especially:

- **Insider attacks**
 - **Intelligent evasion techniques**
-

Let me know if you want a diagram or flowchart to visualize this!